
Fake τ Estimation using GNN

Shahzad Ali, Kathryn Grimm

California State University

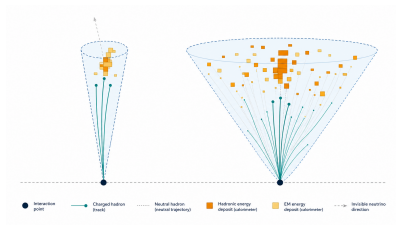
$HH \rightarrow bb\tau\tau$, lep-had (SLT)

August 2026

A jet becomes a fake τ when it passes τ identification

Real $\tau_{\text{had-vis}}$

A real τ decay produces the candidate



Jet $\rightarrow \tau$ fake

A quark/gluon jet produces a τ -like candidate

- ▶ Real hadronic τ decays are narrow, usually with one or three charged tracks
- ▶ A jet can fluctuate to a narrow core and pass the same ID

“Fake” describes the candidate’s origin — not detector noise.

Source: ATLAS, ATL-PHYS-PUB-2019-033. Illustration generated for this presentation.

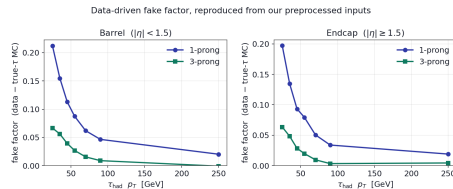
The current method turns anti-ID data into a fake estimate



The binned method reads pre-binned **histograms (ff-inputs)** and looks up a ratio in just three axes $(p_T^{\tau_{\text{had}}}, \eta^{\tau_{\text{had}}}, \text{prong})$, so the weight cannot respond to the rest of the event.

$$\text{FF} = \frac{N_{\text{ID}}}{N_{\text{anti-ID}}} = \frac{(\text{data} - \text{MC})_{\text{ID}}}{(\text{data} - \text{MC})_{\text{anti-ID}}}$$

- ▶ ratio binned in $(p_T^{\tau_{\text{had}}}, \eta^{\tau_{\text{had}}}, \text{prong})$
- ▶ weight each anti-ID candidate to predict the fake yield



FF changes rapidly with $p_T^{\tau_{\text{had}}} / \text{prong}$.

From anti-ID data to an event-level fake prediction



The rest of the talk walks these boxes left to right: H5 inputs $\rightarrow$ soft subtraction $\rightarrow$ training $\rightarrow$ event weight.

What stays the same

Anti-ID data remain the starting point, and simulation is used only to remove real- τ contamination.

What changes

A smooth network replaces the lookup table, so the weight can depend on the full event.

The proposal changes the parameterisation of the Fake Factor — not its data-driven logic.

Step 1 — the H5 inputs reproduce the published Fake Factor

HHARD exports one row per candidate:

- ▶ data and real- τ MC in ID and anti-ID
- ▶ no signal and no MC fake templates
- ▶ event variables plus τ , lepton, jets, MET and MMC objects

$\sim 7\text{M}$ candidates. ID is contaminated by real τ — the next step removes it softly.

First guard: do the H5 inputs reproduce the established binned result?

	our H5	published
inclusive FF	0.0753	0.0754

Agreement to 0.07% — the inputs are sound before any learning.

Step 2 — first isolate fakes softly, then learn their ID odds

1. Soft subtraction

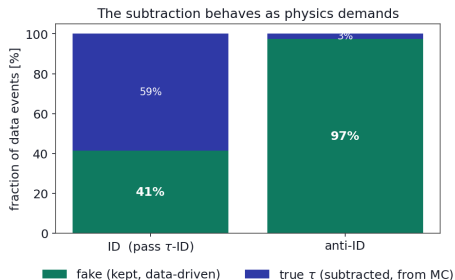
Estimate the real- τ fraction $\rho(x)$ and weight each data event by its fake content

$1 - \rho(x) \in [0, 1]$ — the blue 59% is peeled away.

- ✓ no events discarded in sparse tails
- ✓ weighted totals recover data minus real- τ MC

2. Learn the ratio

On this weighted data, learn the ID-vs-anti-ID ratio; its odds give FF — the next slide makes this precise.



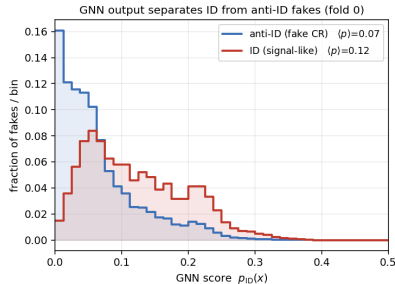
Anti-ID is 97% fake; ID is 59% real τ . Soft subtraction removes the blue by weighting each event by $1 - \rho(x)$, keeping the green fakes.

Step 3 — classifier odds are the local Fake Factor

On the subtracted, fakes-only data, classify:

$$\text{ID} = 1 \quad \text{vs} \quad \text{anti-ID} = 0, \quad \text{FF}(x) = \frac{p(x)}{1 - p(x)}$$

- ▶ $p(x)$: how likely a fake candidate is to pass τ ID
- ▶ natural class proportions $\Rightarrow$ odds = local yield ratio
- ▶ $x = \tau$, lepton, jets, MET + global variables



ID fakes score higher than anti-ID fakes ($\langle p \rangle$ 0.12 vs 0.07): the odds *are* the fake factor.

The target is ID vs anti-ID within the fakes — its odds are the local FF.

Cross-fitting keeps predictions out of sample

- ▶ Split events deterministically into three folds
- ▶ For each rotation: train on one fold, select the checkpoint on a second, and apply only to the untouched third fold
- ▶ Write the cross-fitted soft-subtraction weight $cw(1 - \rho(x))$ once per event
- ▶ Standardisation (norm dict) shared by all three models

Each event is weighted by a model that never saw it.

$$\underbrace{k}_{\text{train}} \rightarrow \underbrace{(k+1)\%3}_{\text{validate}} \rightarrow \underbrace{(k+2)\%3}_{\text{apply}}$$

The same rotation is used for the subtraction model and the ID/anti-ID classifier, preventing information leakage at both stages.

A transformer matches the structure of a collision event

The fake probability depends on both the τ candidate and the event that produced it.

- ▶ A variable number of objects, with no physical ordering
- ▶ Object relationships carry composition information: $\Delta R(\tau, \ell)$, $m_{b\ell}$, $m_T(\tau, \text{MET})$
- ▶ Attention interprets each object in the context of all others

It learns topology without adding histogram axes.

τ candidate lepton b -jets

extra jets MET MMC

↓ self-attention ↓

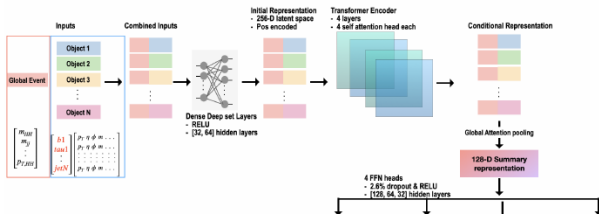
one event representation

↓

$p(\text{ID} \mid \mathbf{x})$

In this analysis, “GNN” refers to this object-based transformer: attention learns the effective connections.

SALT compresses the object set into one ID probability



1. Embed

Object + global features $\rightarrow$ a learned 256-dimensional representation.

2. Exchange context

Four layers $\times$ four attention heads learn object correlations.

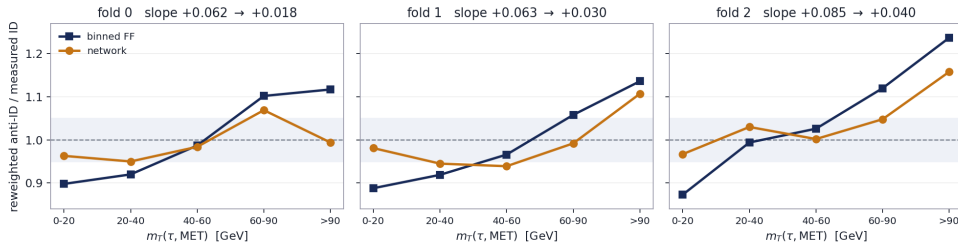
3. Pool and classify

Attention pooling gives a 128-dimensional event summary, then $p(\text{ID} | x)$.

Source: Adapted from the SALT architecture in ATL-COM-PHYS-2025-825, Fig. 7.1; dimensions verified against the fake-factor training configuration.

The binned bias is real — and the network flattens it

The same bias in three independent folds — and the same correction



Closure = (anti-ID data $\times$ weight) / (data – MC)_{ID},
versus m_T^τ — a variable neither method bins. Flat at 1
is perfect closure.

The binned FF (navy) slopes the same way in all three independent folds; the network (amber) reads the event context and stays close to 1.

The same bias appears in three held-out folds — and the network applies the same correction to each.

Held-out data show the gain beyond the binned axes

Both methods use the same train, calibration and application folds. The table reports the RMS of predicted/observed fake yield across bins; lower is better.

variable	binned	GNN	
$p_T^{\tau_{\text{had}}}$ (binned's own axis)	0.13	0.13	tie
m_T^τ (W-veto)	0.105	0.041	✓
$\Delta R(\tau, \ell)$	0.052	0.041	✓
H_T	0.052	0.038	✓
m_{bb}	0.054	0.035	✓
MET	0.058	0.054	✓

The GNN ties the lookup table on $p_T^{\tau_{\text{had}}}$ and improves all five tested composition-sensitive variables.

The clearest change is in m_T^τ : its closure RMS improves from 0.105 to 0.041. The improvement repeats across all three held-out folds.

The learned weight drops into HHARD unchanged

- ▶ Export the three fold models to ONNX
- ▶ Route each anti-ID event to its untouched-fold model
- ▶ Convert the output to $FF(x) = p/(1 - p)$ and use it as the event weight
- ▶ Write learned and binned fakes as separate processes in the same output

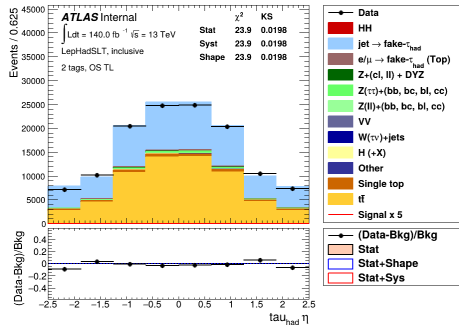
All selections and events are identical; only the anti-ID-to-ID transfer weight changes.

anti-ID data $\xrightarrow{FF_{GNN}(x)}$ fake estimate

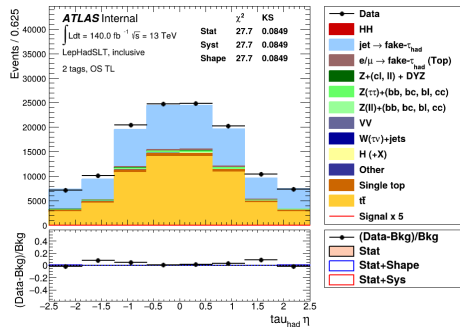
This makes the comparison interpretable: any difference in the prediction comes from the fake-factor parameterisation, not from a different event selection.

The learned estimate preserves the $\tau\eta$ shape

binned Fake Factor



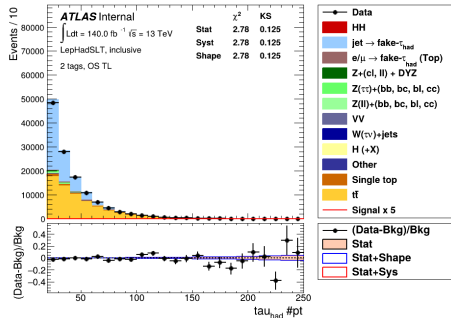
learned (GNN) Fake Factor



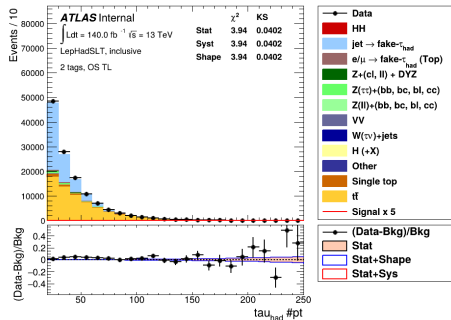
Both describe the $\tau\eta$ shape; the learned fake yield is about 10% lower.

Both methods describe the τp_T spectrum

binned Fake Factor



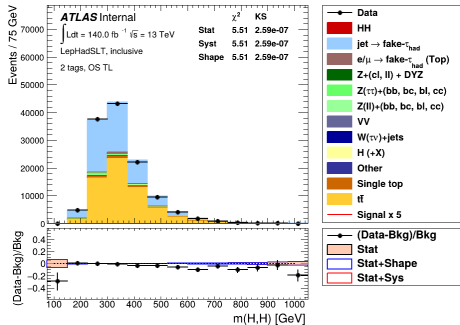
learned (GNN) Fake Factor



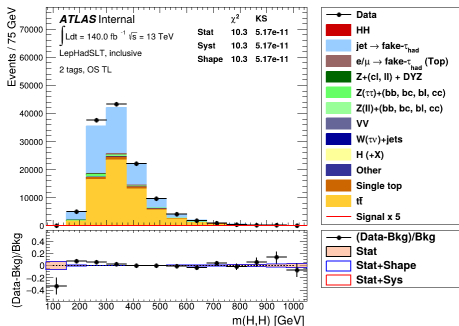
p_T^τ is an axis of the lookup table; the learned method preserves its shape while retaining the overall normalisation shift.

The m_{HH} change appears in the analysis plots

binned Fake Factor



learned (GNN) Fake Factor



The learned weight changes the final discriminant; category-level closure and non-closure uncertainties are therefore essential.

Validation and uncertainties come next

Done

- ✓ three-fold, data-driven event-level Fake Factor
- ✓ improved closure in composition-sensitive variables
- ✓ ONNX deployment in HHARD
- ✓ first side-by-side analysis plots

Next

- ▶ add m_T^τ and m_{bL} to the standard analysis plots
- ▶ test the 10% normalisation shift in dedicated validation regions
- ▶ propagate the estimate to signal-region categories
- ▶ build subtraction, extrapolation and non-closure uncertainties

The event-level Fake Factor runs end to end

- ▶ The standard method transfers anti-ID data with a lookup table in three τ variables
- ▶ SALT learns the same ID/anti-ID ratio continuously from the full event
- ▶ On held-out data it matches the binned method on $p_T^{\tau_{\text{had}}}$ and improves five composition-sensitive variables
- ▶ The ONNX models now run inside HHARD and produce analysis-level fake estimates

The implementation chain is complete; the method becomes analysis-ready only after transport validation and systematic uncertainties.

Thank you — questions welcome.

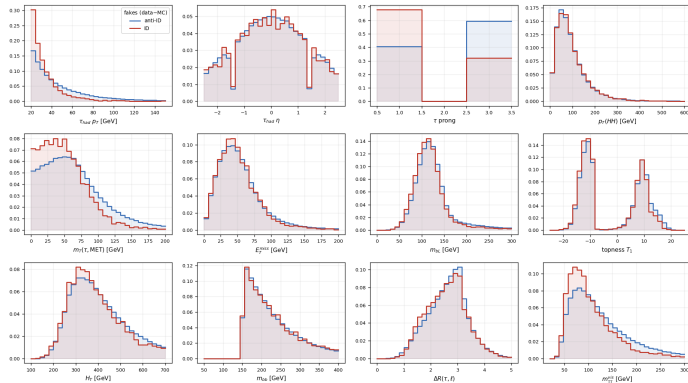


Backup

Training inputs after subtraction

Training inputs after subtraction: ID vs anti-ID

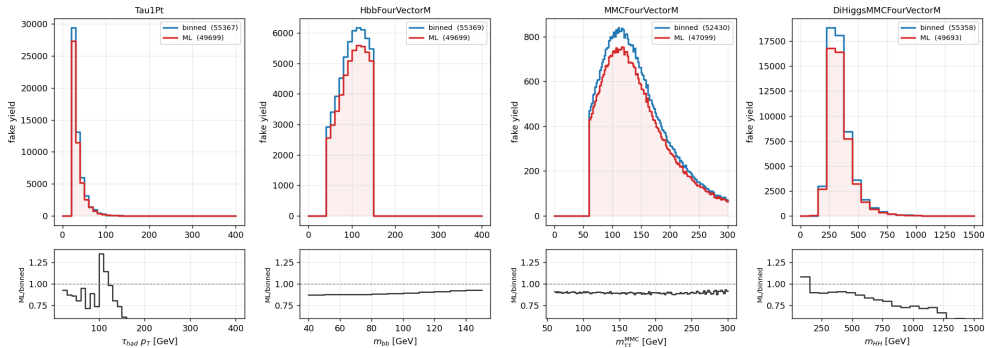
Training inputs after MC subtraction, shape-normalised (fold 0): row 1 = the binned FF's axes + kinematics



The GNN adds composition-sensitive information beyond $p_T/\eta/\text{prong}$.

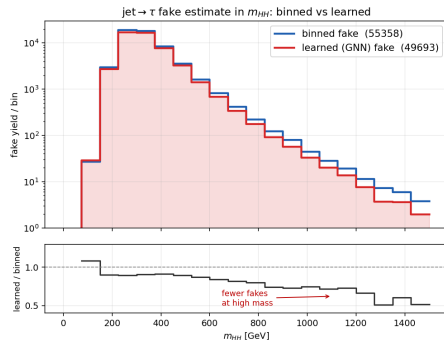
Backup: binned vs learned fake estimate

Fake estimate: binned vs learned (GNN), 2-tag OS anti-ID, Run 2



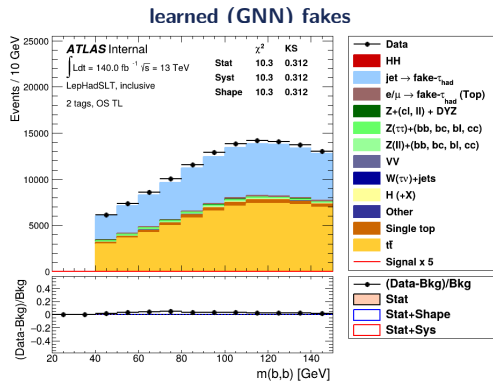
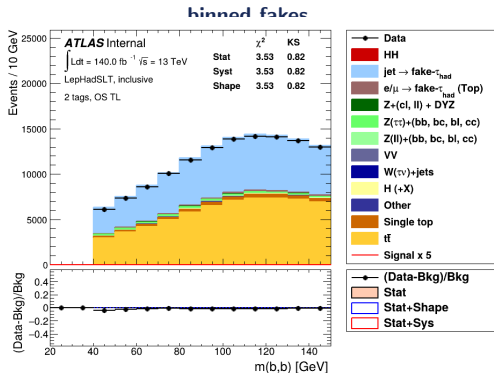
About 10% lower overall, with an additional shape change in m_{HH} .

Backup: the learned weight reshapes m_{HH}

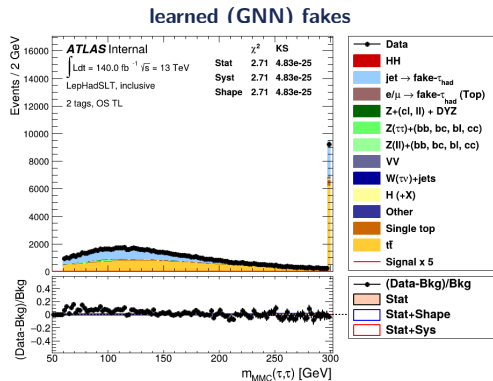
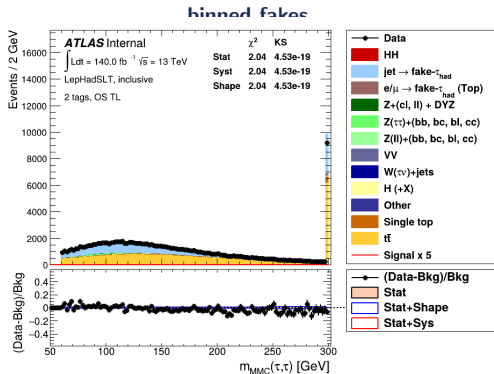


The learned weight predicts fewer fakes toward high m_{HH} — a shape change the binned FF cannot produce.

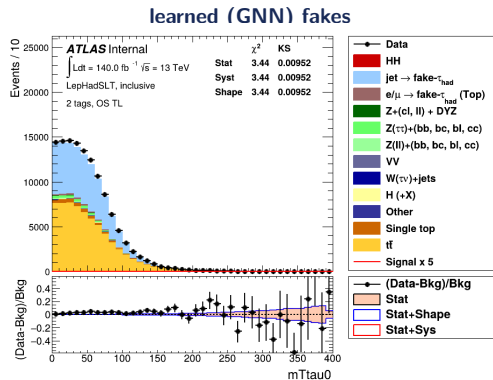
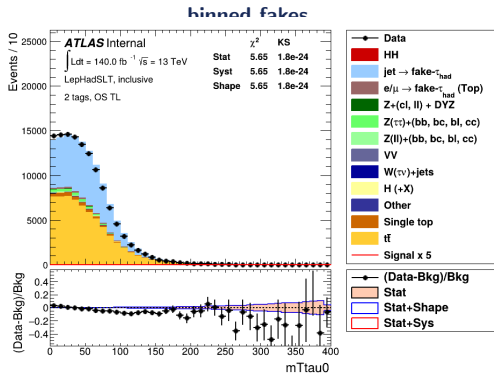
Backup: data/MC in m_{bb}



Backup: data/MC in $m_{\tau\tau}^{\text{MMC}}$

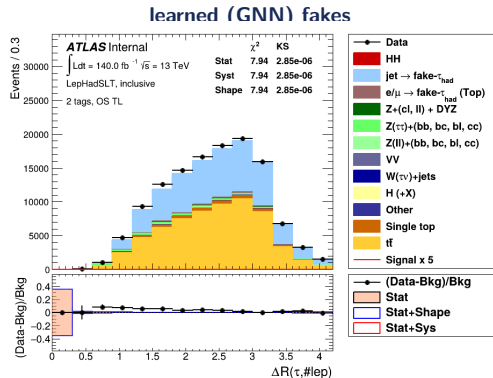
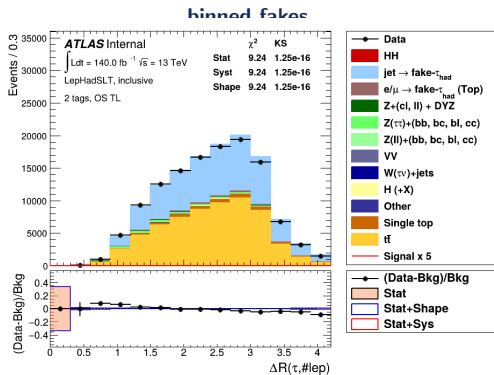


Backup: data/MC in m_T^τ (W-veto)



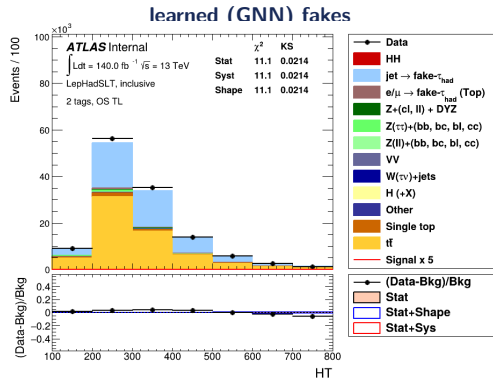
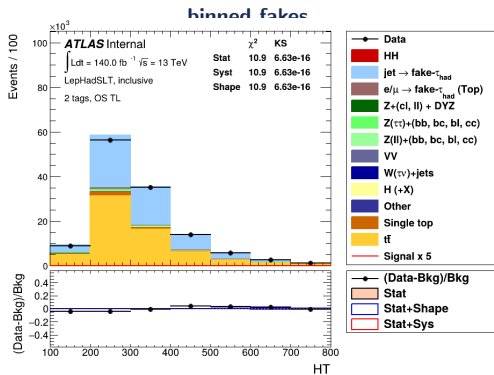
The variable with the largest closure gain (RMS $0.105 \rightarrow 0.041$): the learned fake tracks data where the lookup table sloped off.

Backup: data/MC in $\Delta R(\tau, \ell)$



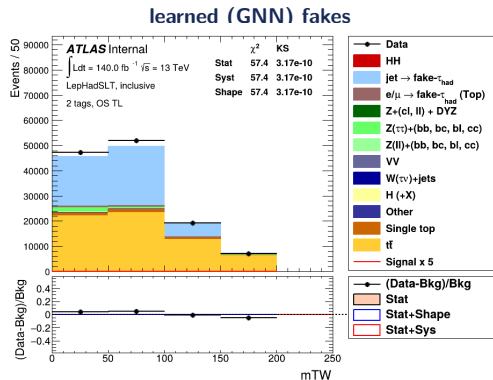
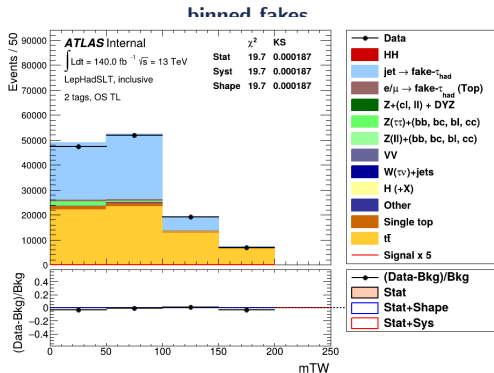
A topology variable the binned axes cannot see (closure RMS $0.052 \rightarrow 0.041$).

Backup: data/MC in H_T



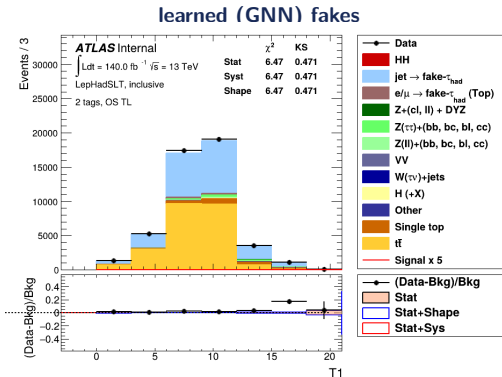
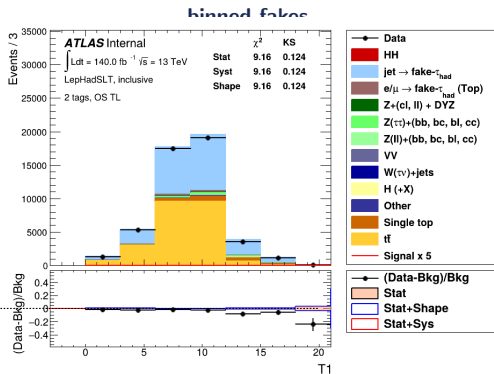
Event-activity dependence the lookup misses (closure RMS $0.052 \rightarrow 0.038$).

Backup: data/MC in m_T^W



Both describe the W -transverse-mass shape; the learned normalisation is slightly lower.

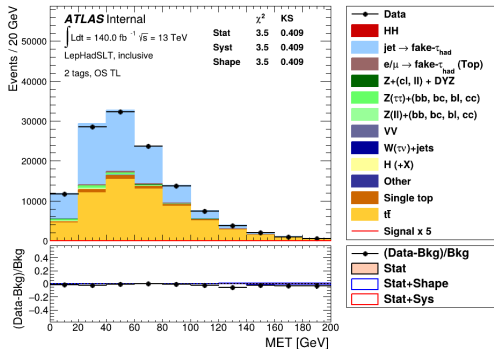
Backup: data/MC in T_1 (thrust)



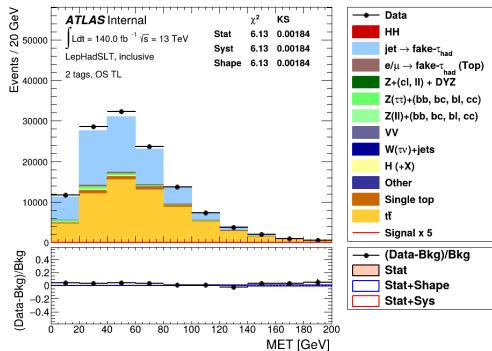
An event-shape variable available to the network but not to the binned axes.

Backup: data/MC in MET

hinned fakes

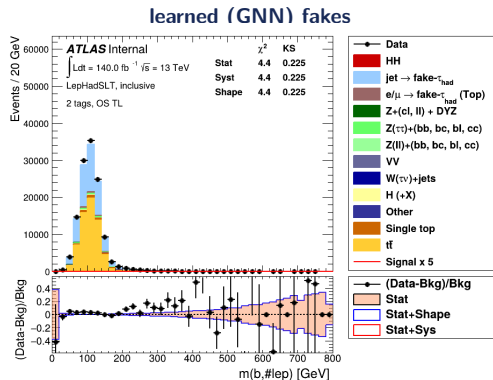
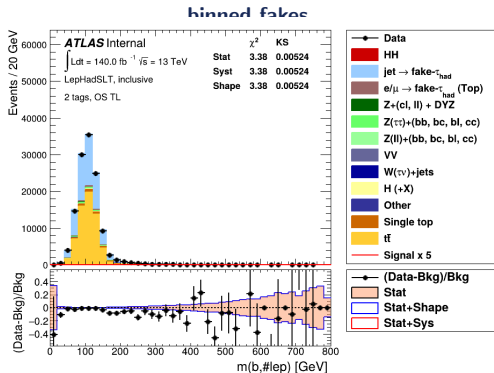


learned (GNN) fakes



MET enters the fake composition (real τ carry neutrinos): closure RMS
 $0.058 \rightarrow 0.054$.

Backup: data/MC in $m_{b\ell}$



The b -lepton mass: another correlated composition variable the lookup axes cannot reach.